a) Define Software Project Management (SPM)

Software Project Management (SPM) is the discipline of **planning, organizing, directing, and controlling** software development activities to ensure that a software project is **completed on time, within budget, and with the required quality**.

It involves managing **people, processes, resources, risks, scope, time, and cost** throughout the software project life cycle.

---

b) Role of a Project Manager

The **Project Manager** is responsible for the **overall success of the software project**.

Key Roles and Responsibilities:

- **Project Planning** – Define scope, objectives, tasks, and schedules
- **Time Management** – Ensure milestones and deadlines are met
- **Cost Management** – Prepare and control the project budget
- **Resource Management** – Allocate and manage human and technical resources
- **Risk Management** – Identify, analyze, and mitigate project risks
- **Communication** – Act as a link between clients, developers, and stakeholders
- **Quality Management** – Ensure deliverables meet required standards
- **Monitoring & Control** – Track progress and take corrective actions

---

c) Explain the Triple Constraint

The **Triple Constraint** (also called the **Project Management Triangle**) represents the **three key factors** that determine the success of a project:

1. Scope

- Defines **what work must be done** and **what features are included**
- Changes in scope affect time and cost

2. Time

- The **schedule** required to complete the project
- Delays increase cost or reduce scope

3. Cost

- The **budget** allocated for the project
- Reducing cost may reduce scope or increase time

Key Concept

All three constraints are **interdependent**:

- Changing one constraint **directly impacts the others**
- A balance is required to maintain **project quality**

Example

If project scope increases →

- More time is needed **or**
- More cost (resources) is required